ÖZYEĞİN UNIVERSITY

PHYS102/Spring 2018-2019 Final Exam

PHYSICS 102 NO: _____

NAME: _____

(On my honor I have neither given nor received any unauthorized assistance on this exam.)

- This exam is out of 110 points, and contains 4 problems.
- This exam is for 100 minutes.
- This is a closed-book/notes/calculator/laptop exam.
- Read the questions carefully and write down your solutions in the space provided neatly and clearly. Write down your final results in the boxes at the end of each question.
- **Answers without work shown or proper explanation will not be given any credit..**

Q1	Q2	Q3	Q4	TOTAL

Formula Sheet

Coulomb's law:

$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2}$

$\frac{1}{4\pi\epsilon_0} = 9\times 10^9\ \mathrm{N\cdot m^2/C^2}$

Electric field:

$\vec{E} = \frac{\vec{F}_0}{q_0}$

$\epsilon_0 = 9\times 10^{-12}\ \mathrm{C^2/N\cdot m^2}$

Electric potential energy:

$W^E_{a\to b} = -\Delta U = U_a - U_b$

$U = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r}$

$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$

Electric flux:

$\Phi_E = \int E\cos\phi\, dA = \int \vec{E}\cdot d\vec{A}$

Gauss's law:

$\Phi_E = \oint \vec{E}\cdot d\vec{A} = \frac{Q_{\mathrm{enc}}}{\epsilon_0}$

Electric potential:

$V = U/q_0$

$V = \frac{1}{4\pi\epsilon_0}\int\frac{dq}{r}$

$V_a - V_b = \int_a^b \vec{E}\cdot d\vec{\ell}$

$E_x = -\frac{\partial V}{\partial x},\ E_y = -\frac{\partial V}{\partial y},\ E_z = -\frac{\partial V}{\partial z}$

$W^E_{a\to b} = -\Delta U = -q\Delta V = -q(V_b - V_a)$

Capacitors and capacitance:

$C = Q/V_{ab}$

$C = \epsilon_0 A/d$

Capacitors in series:

$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$

Capacitors in parallel:

$C_{eq} = C_1 + C_2 + C_3 + \dots$

(W^E is the work done by Electrostatic force.)

$\mu_0 = 4\pi\times 10^{-7}\frac{N}{A^2}$

Capacitors with a dielectric:

$C_\kappa = \kappa C_0\ ,\quad \kappa = \frac{\epsilon}{\epsilon_0}$

Energy in a capacitor:

$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$

Electric energy density:

$u = \frac{1}{2}\epsilon_0 E^2$

Current and current density:

$I = dQ/dt = n|q|v_d A$

$\vec{J} = nq\vec{v}_d$

Resistivity: $\rho = E/J$

Resistors: $V = IR,\quad R = \rho L/A$

Energy and power:

$P = (V_{\mathrm{a}} - V_{\mathrm{b}})I_{\mathrm{b}\to\mathrm{a}}$

$P = I^2R = V^2/R$ (for resistors)

Energy from power:

$U = \int P dt$ (if P is time dependent.)

$U = P(t_f - t_i)$ (if P is time indep.)

Resistors in series:

$R_{eq} = R_1 + R_2 + R_3 + \dots$

Resistors in parallel:

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$

Kirchoff's Rules:

$\sum I = 0$ (junction rule)

$\sum V = 0$ (loop rule)

Charging RC Circuit:

<u>if $Q_0 = 0$:</u> $Q(t) = Q_f(1 - e^{-t/RC})$

<u>if $Q_0 \neq 0$:</u>

$Q(t) = Q_0 + (Q_f - Q_0)(1 - e^{-t/RC})$

$I(t) = I_0 e^{-t/RC},\quad I(t) = \frac{dQ(t)}{dt}$

Discharging capacitor:

$Q(t) = Q_0 e^{-t/RC}$

$I(t) = I_0 e^{-t/RC}$

Time Constant of an RC Circuit:

$\tau = RC$

Magnetic force on a moving charged particle:

$\vec{F}_B = q\vec{v}\times\vec{B}$

Magnetic force on a current carrying conductor:

$\vec{F}_B = I\vec{\ell}\times\vec{B}$

$d\vec{F}_B = Id\vec{\ell}\times\vec{B}$ (on a wire segment.)

Magnetic moment: $\vec{\mu} = I\ \vec{A}$

Magnetic torque: $\vec{\tau} = \vec{\mu}\times\vec{B}$

Potential Energy stored in a current carrying loop placed in B:

$U = -\vec{\mu}.\vec{B}$

Magnetic field of a moving charge:

$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\vec{v}\times\hat{r}}{r^2}$

Magnetic field of a current-carrying conductor: (Biot-Savart Law)

$d\vec{B} = \frac{\mu_0}{4\pi}\frac{Id\vec{\ell}\times\hat{r}}{r^2}$

Magnetic field of a long, straight, current-carrying conductor:

$B = \frac{\mu_0 I}{2\pi R}$

Magnetic force between current-carrying conductors:

$\frac{F}{L} = \frac{\mu_0 I I'}{2\pi R}$

Magnetic fLux:

$\Phi_B = \int \vec{B}\cdot d\vec{A}$

$\Phi_B = \vec{B}\cdot\vec{A}$ (for uniform $\vec{B}$)

Faraday's Law: Induced EMF

$\varepsilon = -\frac{d\Phi_B}{dt}$

Motional EMF

$\varepsilon = \int \vec{v}\times\vec{B}\cdot d\vec{\ell}$

Ampere's Law:

$\oint \vec{B}\cdot d\vec{\ell} = \mu_0 I_{\mathrm{enc}}$

$I_{\mathrm{enc}} = \oint \vec{J}\cdot d\vec{A}$, ($I = \vec{J}\cdot\vec{A}$, if J is constant)

Problem 1 (20 points)

(a) (12 pts) State the four basic equations of electromagnetism, known as Maxwells laws. Give their names.

Mawell equations are:

$$\oint \vec{E}.d\vec{A} = \frac{Q_{enc}}{\epsilon_0} \qquad \text{Gauss' Law} \qquad (1)$$

$$\oint \vec{B}.d\vec{A} = 0 \qquad \text{Gauss' Law for Magnetism} \qquad (2)$$

$$\oint \vec{E}.d\vec{\ell} = -\frac{d\Phi_B}{dt} \qquad \text{Faraday's Law} \qquad (3)$$

$$\oint \vec{B}.d\vec{\ell} = \mu_0 I_{enc} + \mu_0\epsilon_0\frac{\Phi_E}{dt} \qquad \text{Ampere's Law} \qquad (4)$$

(b) (4 pts) Which equations would have to be modified **if magnetic monopoles existed in nature**?

If magnetic monopoles exists, Equation(2) should be:

$$\oint \vec{B}.d\vec{A} = \mu_0 Q_m$$

where Q_m is magnetic charge,

and Equation(3) should be:

$$\oint \vec{E}.d\vec{\ell} = -\mu_0 I_m - \frac{d\Phi_B}{dt}$$

where I_m is magnetic charge current.

(c) (4 pts) What is the physical meaning of the quantity $\frac{1}{\sqrt{\mu_0\epsilon_0}}$?

It is the speed of electromagnetic wave, propagating in free space, and this speed is equal to the speed of light (c).

Problem 2 (30 points)

Total charge Q (positive) is distributed over a region in space. The spherically symmetric charge density is given by:

$$\rho = \alpha, \text{ if } r \leq R,$$
$$\rho = 0, \text{ if } r > R\ ,$$

where α is a constant.

(a) (10 pts) Derive an expression for α in terms of Q and R. Give the unit of α in SI unit system.

$$Q = \int_0^R \rho dV = \int_0^R \alpha dV$$

since α is constant, it can be taken out of the integral sign.

$$Q = \alpha \int_0^R dV = \alpha \frac{4}{3}\pi R^3$$

$$\alpha = \frac{3Q}{4\pi R^3}$$

unit of α in SI is Coulomb/m^3.

(b) (10 pts) Find the electric field for $r \leq R$, and $r > R$.

Using Gauss' Law:

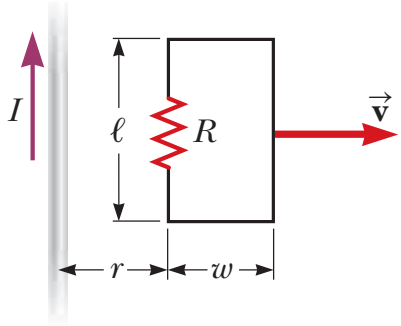
$$\oint \vec{E}.\vec{A} = Q_{enc}/\epsilon_0$$
$$E4\pi r^2 = \alpha \frac{4}{3\epsilon_0}\pi r^3 = Q\frac{r^3}{\epsilon_0 R^3}$$
$$\vec{E}(r) = \frac{Q}{4\pi\epsilon_0}\frac{r}{R^3}\hat{r} \quad \text{for } r \leq R$$

(c) (10 pts) Find the change in the electric potential energy of a positive test charge q, if it moves from $r = R/2$ to $r = R$. Explicitly indicate if the potential energy has increased or decreased.

$$\Delta U_E = q\Delta V$$
$$\Delta V = V(R) - V(R/2) = \int_R^{R/2} \vec{E}(r).d\vec{l} \quad \text{we can choose } d\vec{l} = dr\ \hat{r}$$
$$\Delta V = \int_R^{R/2} \left(\frac{Q}{4\pi\epsilon_0}\frac{r}{R^3}\hat{r}\right)(dr\ \hat{r}) = \frac{Q}{8\pi\epsilon_0}\frac{r^2}{R^3}\Big|_R^{R/2} = -\frac{3}{32\pi\epsilon_0}\frac{Q}{R}$$
$$\Delta U_E = -\frac{3}{32\pi\epsilon_0}\frac{qQ}{R}$$

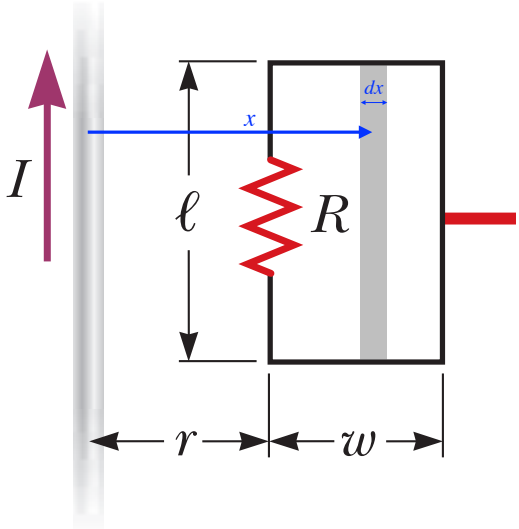
electrical potential energy has decreased.

Problem 3 (30 points)



A rectangular loop of dimensions ℓ and w moves away from a long wire carrying current I, with constant speed v. The total resistance of the loop is R. When the left side of the loop is at a distance r from the wire,

(a) (15 pts) find the magnetic flux through the loop.



The flux through the infinitely small area is:

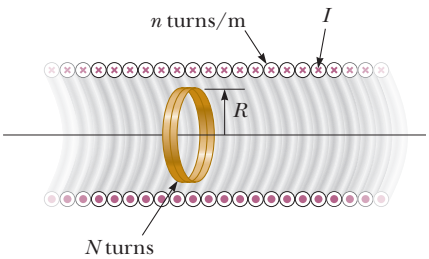
$$d\Phi_B = B(x)dA = \left(\frac{\mu_0 I}{2\pi x}\right)(\ell dx)$$

Integrating from r to $r+\omega$ gives Φ_B:

$$\Phi_B = \int d\Phi_B = \int_r^{r+\omega} \frac{\mu_0 I \ell}{2\pi}\frac{dx}{x} = \frac{\mu_0 I \ell}{2\pi}\ln\left(1+\frac{\omega}{r}\right)$$

(b) (15 pts) Find the magnitude and the direction of the induced current around the loop. (Hint: r is actually $r(t)$).

Problem 4 (30 points)



A long solenoid has n turns per unit length and carries a current $I(t) = I_0(1 - e^{-\alpha t})$. Find the induced electric field at a distance R from the axis of the solenoid.
Hint: Magnetic field inside a long solenoid is given as: $B(t) = \mu_0 n I(t)$